



Guidelines

Thrombolysis for STEMI Medication Guideline

Scope

Site	Service/Department/Unit	Disciplines
SCGH	Coronary Care Unit Emergency Department Intensive Care Unit	Medical, Nursing, Pharmacy

- Patient should be monitored by staff trained in advanced life support and where there is access to a defibrillator.
- Thrombolytics can cause bleeding. Patients must be observed closely during and post administration of their dose.
- Tenecteplase is the preferred thrombolytic for treatment of acute STEMI at SCGH

Introduction

Tenecteplase and Alteplase are recombinant plasminogen activators that are derived from native tPA. They bind to the fibrin component of the thrombus and converts thrombus bound plasminogen to plasmin, which degrades the fibrin matrix of the thrombus.⁽¹⁾

1.0 Presentation³

Tenecteplase 50 mg (10,000 units) vial, with 10 mL water for injection diluent prefilled syringe (*Metalyse*®) or 10mL vial of water for injections (*TNKase*®).

Alteplase 10mg vial with 10ml diluent vial of water for injections, and 50mg vial with 50mL diluent vial of water for injections.

2.0 Indications^{1,2,4}

Acute STEMI

- Tenecteplase and alteplase are indicated for the thrombolytic treatment of acute ST Elevation Myocardial Infarction (STEMI), when timely access to percutaneous coronary intervention is not available (within 120 minutes of first medical contact) and at the direction of a cardiologist.
- Treatment must be given within 12 hours of symptom onset of STEMI.
- Dual antiplatelet therapy (Aspirin 300mg and Clopidogrel 300mg), plus weight adjusted heparin or enoxaparin is routinely given – see section 8 for dosing.

3.0 Contraindications^{1,2}

Must be completed

Absolute	YES	NO
Known allergy/hypersensitivity reaction to thrombolytic agents or gentamicin		
Active or recent internal bleeding <14 days (excluding menstruation)		
Significant closed head trauma, facial trauma or other severe trauma within past 3 months		
Suspected aortic dissection or pericarditis		
Any prior intracranial haemorrhage or previous haemorrhagic stroke		
Ischaemic stroke within the past 12 months		
Known structural cerebral vascular lesion (e.g. arteriovenous malformation, aneurysm)		
Known malignant intracranial or intraspinal neoplasm (primary or metastatic)		
Known severe bleeding disorder		
Recent (within 2 months) intracranial or intraspinal surgery		
Relative	YES	NO
Current use of oral anticoagulants		
Recent major surgery (< 3 weeks)		
Organ biopsy or non-compressible vascular punctures <10 days		
Traumatic or prolonged (> 10 minutes) CPR		
Recent internal bleeding (2-4 weeks prior)		
Gastrointestinal bleed within 1 month		
Severe thrombocytopenia (platelet count <50 or evidence of petechiae/bruising)		
History of chronic, severe, poorly controlled hypertension or severe uncontrolled hypertension on presentation (SBP >180mmHg or DBP >110mmHg)		
Dementia or known intracranial pathology (e.g. arteriovenous malformation, brain tumour, previous stroke)		
Pregnancy		
Haemorrhagic ophthalmic conditions		
Active peptic ulcer during the last 3 months		
Severe hepatic or renal dysfunction		
Infective endocarditis		
If YES to any question(s) consult with a cardiologist for advice		

4.0 Pre-administration^{5,6}

- Obtain informed verbal consent and document in medical record – see patient information sheet
- Attach to cardiac monitor and obtain and record vital signs.
- Obtain derived 12-lead ECG.
- Establish IV access x2
- Collect blood for FBP, U&E, troponin, BSL, Coags, Group and Hold.
- Apply pressure dressings to puncture sites and superficial wounds
- Consider pain relief and anti-emetics as required
- Administer Aspirin 300mg PO and Clopidogrel 300mg PO

- Consider IV Glyceryl Trinitrate infusion for SBP >180mmHg or DBP >110mmHg.
- Give unfractionated heparin (**preferred**) or enoxaparin bolus (Refer to Anticoagulation with Thrombolytic Therapy section).
 - Unfractionated heparin **bolus** 60iu/kg up to 4000 units (followed by heparin infusion).

OR

- Enoxaparin 30mg IV **bolus** if <75 yrs (followed by dosing based on weight and renal function)

Note **DO NOT** give enoxaparin IV bolus if patient >75 yrs.

5.0 Reconstitution³

5.1 Tenecteplase - Metalyse

- Remove the cap from the prefilled syringe and screw onto the vial adaptor.
- Push the spike of the vial adaptor into the vial and slowly inject the diluent.
- Swirl gently to dissolve.¹ **Do not shake.**

5.2 Tenecteplase – TNKase

- Use the syringe supplied to withdraw 10 mL of water for injections from the diluent vial and use this to reconstitute the TNKase vial.
- Swirl the vial gently until completely dissolved. **Do not shake.**
- Slight foaming will subside in a few minutes

5.3 Alteplase

- Reconstitute the 10 mg vial with 10 mL and the 50 mg vial with 50 mL of water for injections to make a concentration of 1 mg/mL. Use the transfer cannula provided or a large bore (18 gauge) needle.
- Swirl gently until completely dissolved. **Do not shake.**
- Allow to stand for several minutes to allow bubbles to dissipate.

6.0 Dose & Administration^{1,2,4,5,6}

Tenecteplase is the preferred thrombolytic for the treatment of acute STEMI at SCGH.

The dose of Tenecteplase and Alteplase used for the treatment of acute STEMI is different to the dose used for acute ischaemic stroke

6.1 Tenecteplase:

- Dose is calculated based on bodyweight, with a maximum dose of 50 mg (10,000 units). See dose table below.^{1,3} The dose to be administered should be expressed in milligrams.
- Dose is administered as a single intravenous (IV) bolus over 10 seconds. Flush the line before and after injection to avoid precipitation in the line.²
- **Caution with patients ≥ 75 years:** consider using half the recommended dose due to a higher bleeding risk (intracranial haemorrhage).³

Weight (kg)	Dose in units	Dose in mg	Volume in mL
≤ 60 kg	6000 units	30 mg	6 mL
60-69 kg	7000 units	35 mg	7 mL
70-79 kg	8000 units	40 mg	8 mL
80-89 kg	9000 units	45 mg	9 mL
≥ 90 kg	10,000 units	50 mg	10 mL

6.2 Alteplase:

- Less than 65 kg: 15 mg intravenously (over 2 minutes), followed by 0.75 mg/kg by intravenous infusion over 30 minutes, then 0.5 mg/kg over the next 60 minutes
- 65 kg or more: 15 mg intravenously (over 2 minutes), followed by 50 mg by intravenous infusion over 30 minutes, then 35 mg over the next 60 minutes

7.0 Anticoagulation with Thrombolytic Therapy^{5,6}

- Parenteral anticoagulation must be given with thrombolytic therapy.
- Parenteral anticoagulation should be given until revascularisation (if performed), or at least 48 hours.
- Give unfractionated heparin (**preferred at SCGH**) or enoxaparin.

7.1 Unfractionated heparin

- As per the Acute Coronary Syndrome nomogram on the WA Anticoagulation Medication Chart.

Dilution	Bolus dose	Infusion Rate	Monitoring
25,000 units in normal saline 0.9% 500mL	60 units/kg (up to 4000 units)	12units/kg/hour (up to 1000 units/hour – 20 mL/hour)	Adjust according to APTT

7.2 Enoxaparin

Bolus dose	Maintenance dose – give 15 minutes after bolus		
	Age < 75yrs	Age ≥ 75yrs	Renal impairment (Cr/CL <30mL/min)
Enoxaparin 30mg IV if <75 yrs (Do not give bolus if >75 yrs)	1mg/kg subcutaneously BD (Maximum of 100mg for first 2 doses)	0.75mg/kg subcutaneously BD (Maximum for first 2 doses)	Use unfractionated heparin

- Ensure patients have been administered a loading dose of dual antiplatelet therapy - Aspirin 300mg and Clopidogrel 300mg.
(Note there is limited data on using prasugrel and ticagrelor at the time of thrombolysis. Patients can be switched to prasugrel or ticagrelor 12 to 24 hours after thrombolysis).

8.0 Monitoring^{5,6}

- Patient should be monitored in a critical care area during and post administration of thrombolytic, observing for reperfusion arrhythmias and hypotension.
- Diagnostic and Derived 12 Lead ECG - Immediately post thrombolysis, 6 hours post thrombolysis, 24 hours thrombolysis and then daily while in hospital.
- Troponin - Immediately post thrombolysis, 6 hours post thrombolysis and then daily for three days.
- Plasma fibrinogen levels should be checked immediately post thrombolysis and if invasive procedures are planned. These should be deferred if the plasma fibrinogen is <0.2G/L.
- APTT to be monitored as per the WA Anticoagulation Medication Chart:
 - at baseline, within 6 hours of commencing infusion and within 6 hours of each rate change.
- Obtain and record vital signs and pain score 15 minutely for one hour, hourly for two hours and four hourly if stable.
- For significant signs and symptoms of bleeding from intracranial, gastrointestinal genitourinary, gingival and retroperitoneal sites.
- For signs and symptoms of bleeding from vascular puncture and access sites.
- No venepuncture or intramuscular injections within 24 hours of Tenecteplase administration. If bleeding occurs, apply pressure to site for 30 minutes and check frequently.
- In the case of severe bleeding not controlled by pressure, stop the infusion of Heparin and consider use of fibrinogen, platelets, coagulation factors or tranexamic acid.^{1,3}

9.0 Drug Interactions^{1,2}

- Concomitant use of anticoagulants or antiplatelets may increase the risk of bleeding prior to, during or after thrombolytic therapy.
- The concomitant use of Glycoprotein IIb/ IIIb receptor antagonists such as tirofiban and eptifibatide increases the risk of bleeding.

10.0 Adverse Effects^{1,2}

- Bleeding at puncture site and haematoma are common.
- Significant bleeding from intracranial, gastrointestinal, genitourinary and retroperitoneal sites can occur.
- Reperfusion arrhythmias
- Transient hypotension.
- Nausea and vomiting.
- Headache.
- Fever, chills or rash.
- Bronchospasm
- Anaphylaxis
 - Alteplase is associated with an increased risk of angioedema, particularly in patients concurrently taking ACE inhibitors. Angioedema occurs most frequently in the first 2 hours but can occur up to 24 hours post administration.

11.0 Compatibility³

11.1 Tenecteplase

- Fluids: Sodium Chloride 0.9%, Water for injections.
- Y-site: Not recommended.
- Incompatible with glucose 5%

11.2 Alteplase

- Fluids: Sodium chloride 0.9%
- Y-site: Eptifibatide, lidocaine, metoprolol
- Incompatible with glucose 5%, Bivalirudin, dobutamine, dopamine, glyceryl trinitrate, heparin sodium

12.0 Pregnancy and Lactation^{1,2}

- Pregnancy: Limited human data; only use if necessary.
Tenecteplase ADEC pregnancy category C.
Alteplase ADEC pregnancy category B¹
- Breastfeeding: No data available but unlikely to be absorbed by the infant.

13.0 Stability³

13.1 Tenecteplase

- Vial: Store below 30 °C. Protect from light.
- Metalyse reconstituted solution: Stable for 8 hours at 30 °C or for 24 hours at 2 to 8 °C.
- TNKase reconstituted solution: use immediately or stable for 8 hours at 2 to 8 °C

13.2 Alteplase

- Vial: store below 30 °C. Protect from light.
- Actilyse reconstituted solution: use immediately or stable for 24 hours at 2 to 8 °C.
- Activase reconstituted solution: use immediately or stable for 24 hours at 2 to 8 °C and for 8 hours below 25 °C.

Related Legislation and Policy Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising policy and standards:

- [SCGH Medicines Management Policy](#)
- [SCGH High Risk Medication Policy](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)
- [WACHS MR 172A](#)
- NSQHS Standards 1, 4

References

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

Patient Information Sheet

Cardiac Thrombolysis – Tenecteplase

We think you are having a heart attack due to your symptoms, the physical examination by the nurse or doctor and from the results of your electrocardiograph (ECG).

Heart attacks are usually due to a blood clot in an artery in the heart. This stops the blood carrying oxygen reaching part of the heart muscle. The blockage causes pain and will probably result in permanent damage to your heart if not treated straight away.

Your ECG has been reviewed by a doctor who has then prescribed 'thrombolysis'. This is the use of a medication to dissolve the clot. In WA public hospitals the medication used is Tenecteplase.

What is my proposed treatment?

Thrombolysis (Tenecteplase) is a drug given by intravenous infusion (a 'drip') to dissolve the blood clot blocking an artery. The nurse may give you the drug before you see a doctor. The sooner you receive it after your symptoms start, the greater the benefit and the less permanent damage to your heart muscle.

Are there any risks?

Thrombolysis is generally a safe procedure but as with most medical procedures there can be risks and complications. To reduce the risks, the doctor/nurse will ask you questions about any existing health conditions and recent surgery.

The key risk of this treatment is possible bleeding either in the brain or from other organs*:

- Minor bleeding can happen for 1 in 4 patients and excessive bleeding can occur in 5% of cases
- Bleeding in the brain (stroke) can happen in 1% of patients under 75 years old. In patients over 75 the risk is approximately 2%.

These risks are usually resolved by stopping treatment or giving a blood transfusion. For most patients, the risks from not treating a blocked artery in your heart are greater than the risks of bleeding elsewhere in your body.

What Happens Now?

The nurse or doctor will discuss further treatment options with you after giving you the thrombolysis. This can include further treatments. If you have any concerns about your treatment, please discuss with the doctor/nurse looking after you.

* Figures quoted are for Tenecteplase (Metalyse)*